JOSEPH YANNEY-EWUSIE NEWTON

Climate Data | Environmental & Geospatial Data Analyst | Carbon MRV
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LinkedIn | GitHub | Website

PROFESSIONAL SUMMARY

Environmental data analyst who combines climate, agriculture, and sustainability expertise with a broad analytical toolkit, spanning GHG accounting, climate and environmental data analysis, and geospatial and remote sensing. Builds end-to-end data pipelines that turn raw data into defensible, decision-ready outputs. MSc in Environmental Sustainability (TH Bingen) and BSc in Agricultural Engineering, with in-depth knowledge of West African agricultural supply chains and experience across three continents in AgriTech, research, and logistics. Known for rigorous data quality, and clear communication in multicultural teams.

TECHNICAL SKILLS

Geospatial & Remote Sensing: Google Earth Engine, Sentinel-2, Landsat, land-cover classification (Random Forest), change detection, QGIS, raster/vector analysis (GeoTIFF, GeoJSON).

Programming & Analysis: Python (geospatial: rasterio, numpy, pandas; ETL, time-series), SQL.

Carbon & Compliance: Carbon MRV, REDD+ / Verra methodologies (VM0048), EUDR deforestation screening, IPCC emission factors, uncertainty analysis, GHG accounting, Net Zero frameworks, Nature-Based Solutions.

Data & Visualization: PostgreSQL, Power BI, Tableau, Streamlit, Excel (Advanced), Matplotlib

Tools & Collaboration: Git / GitHub, AI-assisted development (Claude, Claude Code), Jira, Confluence, Microsoft 365, PowerPoint.

Core Competencies: Cross-functional stakeholder coordination, project ownership and follow-through, translating technical analysis into decision-support, strategic and systems thinking, adaptability in international settings.

SELECTED PROJECTS

Forest Carbon MRV Prototype: Kakum National Park, Ghana [GitHub]

- Built a forest carbon MRV pipeline structured to Verra REDD+ (VM0048), estimating ≈12.5M tCO_2e (≈4.8M conservative, ±61.5%) from Sentinel-2, using Random Forest classification in GEE (95.5% accuracy) and a tested Python package.
- Identified the biomass default, not classification, as the dominant error source, and prioritized a GEDI-based Tier 2 biomass upgrade.

GHG Inventory Tool (Scope 1 & 2) | Live demo [GitHub]

- Built a Python GHG Protocol Scope 1 and 2 inventory tool with per-gas calculations, dual Scope 2 reporting, uncertainty analysis, and audit-ready PDF and Excel outputs.
- Added an abatement analysis that ranks seven reduction options by € per tonne and deployed the tool as a Streamlit app for data editing, scenarios and report export.

EUDR Plot-Level Deforestation Screening (Cocoa, Ghana) [GitHub]

- Built a tool that screens farm plots for post-2020 forest loss (Hansen GFC) in GEE and produces DDS-style risk reports through a tested, config-driven Python module.
- Designed it commodity-agnostic (demonstrated on Ghanaian cocoa, extends to rubber, coffee, palm) and documented why shaded cocoa looks like forest from space, the reason EUDR requires operator GPS polygons.

Bingen Green Roof Climate Data Pipeline & Analytics Dashboard [GitHub]

- Built a production ETL pipeline that ingests, QC-filters, harmonizes, and synchronizes multi-vendor environmental sensor data into a 2.18M-record time-series warehouse (2020–2026), with incremental ingestion and atomic zero-downtime deployment.
- Developed an interactive Streamlit dashboard with statistical significance testing and energy-balance analysis to support research on green roof cooling performance.

PROFESSIONAL EXPERIENCE

Werkstudent | HTWest GmbH — Apr 2026 – Date
Bonn, Germany

- Built a Python GHG Protocol Scope 1 and 2 inventory tool with per-gas calculations, dual Scope 2 reporting, uncertainty analysis, and audit-ready PDF and Excel outputs.
- Assisted in the production, packaging and transportation of bespoke cutting of specialty papers such as parafilm, filter papers, wrappers, etc

Werkstudent, Data Quality | BASF Digital Farming GmbH (xarvio) Oct 2024 – Feb 2026
Cologne, Germany

- Performed structured data quality validation and documentation for digital farming datasets, supporting reliable model outputs and operational decisions.
- Coordinated with teams in Germany, Argentina, and Brazil to resolve data quality issues and align on process standards across functions.
- Produced stakeholder summaries and status updates that improved transparency and follow-through across project cycles.
- Contributed to continuous improvement by tracking recurring quality issues and supporting corrective actions.

University Research Assistant | Technische Hochschule Bingen Sep 2023 – Dec 2023
Bingen am Rhein, Germany

- Collected, validated, and structured field sensor data for climate adaptation research on crop moisture stress and temperature impacts.
- Ensured data integrity and traceability across monitoring locations through consistent documentation and quality checks.
- Presented interim findings to academic supervisors and stakeholders, supporting evidence-based project direction.

Project Analyst | Freezelink Cold Chain Logistics Jan 2022 – Aug 2022
Accra, Ghana

- Compiled benchmark reports on cold chain performance across client sites, informing implementation and operational improvement decisions.
- Supported planning and rollout of modular cooling solutions through cross-team information coordination.
- Maintained structured project documentation and produced efficiency reports used in client reviews.

Trainee | Tzafarti Ibudim Dec 2020 – Oct 2021
Moshav Ahituv, Israel

- Led and coordinated a multicultural team during peak operations, supporting a 95% product acceptance rate through structured quality oversight.

EDUCATION

MSc, Environmental Sustainability | Technische Hochschule Bingen
Thesis: Green Roof Cooling Effectiveness Analysis (urban climate adaptation, environmental data pipeline design)

BSc, Agricultural Engineering | University of Ghana
Accra, Ghana

Diploma in Agriculture (Horticulture) | AgroStudies Training Center
Netanya, Israel

CERTIFICATIONS

- Google Data Analytics Professional Certificate - Coursera (R, SQL, Tableau, Excel)
- BCG Climate & Sustainability Job Simulation - Forage
- OPF Academy (Cohort 6) - Net Zero Strategies, GHG Accounting, Sustainability Frameworks

ADDITIONAL EXPERIENCE

Volunteer Data Analyst | Net Zero Tracker

- Monitors and documents corporate and national net zero commitments across 50+ entities, producing structured summaries for sustainability reporting.

Environmental Volunteer | Service Civil International (SCI Germany)
Upper Middle Rhine Valley

- Supported landscape restoration, building wooden paths and dry-stone walls to prevent erosion in vineyard terrain.

LANGUAGES

English — Fluent · German — A2 (Actively Learning)